The Paris Targets as Numbers

> Part of the Functional Series. The Agreement states its goals as temperatures. To compare them with emissions data, they have to be turned into quantities: how much can still be emitted, how fast emissions must fall, and by when. This document does that translation and lays the platform's pathways next to it. Where the platform's own numbers are used, they come from the scenarios described in Scenarios and Temperature Mapping.
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> **As of:** IPCC AR6 (2021–2023); Forster et al. (June 2026) and the Global Carbon Budget 2025 for budgets; UNEP *Emissions Gap Report 2025* for 2035 benchmarks; the platform's data to 2024.

1. The goals in numbers

Goal	In numbers	Source
Temperature	Well below 2 °C, pursue 1.5 °C, above 1850–1900	Paris Agreement, Article 2.1(a)
Where warming stands	About 1.4 °C long-term (observed 2023–2025 average about 1.5 °C)	Copernicus; Forster et al. 2026 (human-induced 1.37 °C in 2025)
Emissions in 2030	GHG –43% (range –34 to –60%) and CO_2 –48% from 2019 for 1.5 °C; GHG –21% and CO_2 –22% for 2 °C	IPCC AR6 Synthesis Report, Table SPM.1
Emissions in 2035	GHG –60% and CO_2 –65% (1.5 °C); GHG –35% and CO_2 –37% (2 °C)	IPCC AR6 Synthesis Report, Table SPM.1
2035 in UNEP's pathway set	–55% (1.5 °C) and –35% (2 °C) in total GHG	UNEP *Emissions Gap Report 2025*
Net zero CO_2	Early 2050s (1.5 °C); early 2070s (2 °C)	IPCC AR6 Synthesis Report B.6.1
Net zero GHG	Around the 2070s in 1.5 °C pathways that reach it	IPCC AR6 Synthesis Report B.6.1
Remaining carbon budget, 1.5 °C (50%)	130 $GtCO_2$ (Forster et al.) or 170 $GtCO_2$ (Global Carbon Budget), from the start of 2026	See Benchmarks
Remaining carbon budget, 2 °C (50%)	about 1,050 $GtCO_2$, from the start of 2026	Forster et al.; Global Carbon Budget 1,055
Peak	GHG emissions peak between 2020 and "at the latest before 2025" in 1.5 °C pathways	IPCC AR6 Synthesis Report B.6.1

The 43% and 60% figures are the numbers the first global stocktake reaffirmed in 2023.

2. How fast would emissions have to fall?

The pathway figures are reductions from 2019. Observed emissions have since risen (fossil CO_2 was 4.1% above 2019 in 2024), so the required rate from today is steeper than the headline percentage suggests. The table converts each CO_2 pathway into the **constant annual percentage cut from the 2024 level** that would reach it:

Pathway (CO_2, from 2019)	By 2030	By 2035	By 2040
1.5 °C median reduction	–48%	–65%	–80%
Constant cut needed each year from 2024	**10.9% a year**	9.4% a year	9.8% a year
2 °C median reduction	–22%	–37%	–51%
Constant cut needed each year from 2024	**4.7% a year**	4.5% a year	4.6% a year

(Calculated on fossil CO_2 including bunkers, 38.6 $GtCO_2$ in 2024 and 37.1 in 2019. Including land use changes the rates by under 0.3 percentage points.)

For scale:

- **The largest annual fall since 1960 was in 2020, the pandemic year: –5.2%** in the platform's fossil CO_2 series (the IPCC gives –5.8% for CO_2 from fossil fuels and industry). The next largest falls were about –3.0% (1992) and –2.7% (1981).
- A **sustained** 4.5–4.7% a year, the 2 °C requirement, is roughly the 2020 pandemic fall repeated every year for 15 years. The 1.5 °C requirement of about 10% a year is roughly twice that.
- These are **median pathway** requirements. The ranges in the IPCC's table are wide (for the 1.5 °C 2030 CO_2 reduction, –36% to –69%).

3. The platform's pathways against the targets

The platform's scenarios (CO_2, 2025–2040) are shown in Pathways. In summary, by 2035 relative to 2019:

	BAU	Moderate (–2%/yr)	Aggressive (–5%/yr)	IPCC 2 °C median	IPCC 1.5 °C median
CO_2 change by 2035	+16%	–5%	–31%	–37%	–65%

No platform scenario meets even the 2 °C median pathway. The platform's scenarios are illustrative bounds, not forecasts of policy, but the arithmetic is informative. Because the cuts are applied to a business-as-usual path that is still growing, the Aggressive scenario's 5% cut is a net fall of only about 3.8% a year between 2024 and 2040 (38.6 to 20.8 $GtCO_2$). Sustained for 16 years, that is faster than every single year in the past 60 except the 2020 pandemic, and it is still short of the roughly 4.6% a year the 2 °C median requires.

4. When would the carbon budgets run out?

Cumulative CO_2 from the platform's scenarios (fossil and cement, plus land use held flat at about 4.7 $GtCO_2$ a year) can be run against the remaining budgets. Each cell is the first year the scenario's cumulative emissions since the start of 2026 reach the budget, at a 50% likelihood.

Budget from the start of 2026 (Forster et al. / Global Carbon Budget)	BAU	Moderate	Aggressive
1.5 °C: 130 / 170 $GtCO_2$	2028 / 2029	2029 / 2029	2029 / 2030

Budget from the start of 2026 (Forster et al. / Global Carbon Budget)	BAU	Moderate	Aggressive
1.7 °C: 500 / 525 $GtCO_2$	2036 / 2037	2038 / 2038	not by 2040 (496 emitted by 2040)
2 °C: 1,050 / 1,055 $GtCO_2$	not by 2040	not by 2040	not by 2040

(Each year pair reads the result under Forster et al.'s budget / the Global Carbon Budget's.) Cumulative emissions from 2026 to 2040 are 702 $GtCO_2$ for BAU, 608 for Moderate and 496 for Aggressive.

How to read this:

- **The 1.5 °C budget is used up around 2028–2030 in every scenario**, whichever of the two budget estimates is used. The difference between the 130 and 170 figures moves the date by at most a year, and the scenarios themselves differ by at most about two years (2028 for BAU, 2030 for Aggressive): the budget is consumed faster than any of these pathways can bend.
- **The 1.7 °C budget is passed in the late 2030s except under the Aggressive scenario**, which reaches it only at the end of 2040.
- **The 2 °C budget lasts beyond 2040 in all three.**
- **Accounting basis.** The platform's annual CO_2 is gross of the roughly 0.7 $GtCO_2$ cement carbonation sink that the Global Carbon Budget nets out, and its land-use assumption (4.66 $GtCO_2$ a year, flat) is above the Global Carbon Budget's 4.1 $GtCO_2$ for 2025. Together that puts the platform's cumulative emissions about 1.3 $GtCO_2$ a year above the basis the budgets use, so each crossing year comes out slightly early.
- **Using up the budget is not the same as reaching 1.5 °C.** The published budgets include uncertainty in how warming responds to CO_2 only, assume a further 0.10 °C of future non-CO_2 warming, and are for a 50% chance. The overshoot the COP30 decision acknowledged is related: the budget is spent before emissions reach zero.

5. What the targets imply for the platform's own temperature path

The platform's BAU temperature path reaches a level of 1.5 °C in 2033 and Aggressive in 2034 (see Pathways §5). That is later than the budgets run out in §4 (2028–2030), and the reason is specific. The two start from almost the same level: the platform's implied level at the start of 2026 is 1.34 °C, and Forster et al.'s 130 $GtCO_2$ budget implies about 1.34 °C too (1.5 – 0.10 – 130 × 0.45 per 1,000 $GtCO_2$). The difference is Forster et al.'s allowance of 0.10 °C for future warming from non-CO_2 gases, worth about 220 $GtCO_2$ at that slope, which the platform's CO_2-only slope does not include. Without it the platform's room to 1.5 °C from the start of 2026 is about 330 $GtCO_2$ (0.16 °C ÷ 0.485), against Forster et al.'s 130. That is consistent with the IPCC's finding that the first 20-year average to reach 1.5 °C is centred in the first half of the 2030s in all but its highest-emissions scenario, and with Forster et al.'s estimate of around 2030 for human-induced warming. The Paris goal concerns the long-term level, so a single warm year above 1.5 °C (2024 already was) does not by itself mean the goal is breached.

See also

- The Paris Agreement and the COP Process
- Country NDCs
- Scenarios and Temperature Mapping